

Project Design Phase Proposed Solution Template

Date	20/01/26
Team ID	LTVIP2026TMIDS76471
Project Name	Exploratory Analysis of Rainfall Data in India for Agriculture
Maximum Marks	2 Marks

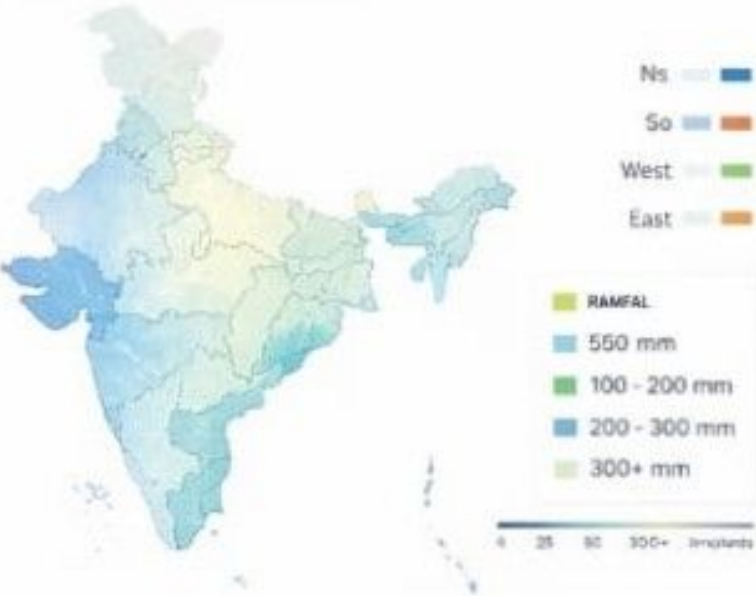
Proposed Solution Template:

S.No.	Parameter	Description
1	Problem Statement	The real estate market involves vast and complex datasets on housing features and sale prices. These datasets are often underutilized due to lack of effective visualization, making it difficult for buyers, sellers, and analysts to draw insights or forecast trends.
2		Our solution transforms static housing datasets into interactive, insightful visualizations using Tableau. The project involves cleaning and transforming the data, creating calculated fields and KPIs, and developing a dashboard that highlights key trends, comparisons, and location-based analyses. The solution is deployed via a Flask web app.
3	Novelty / Uniqueness	This project leverages Tableau's powerful visual capabilities to go beyond basic data analytics. By combining calculated fields, condition segmentation, and geographic mapping, the dashboard offers a dynamic exploration of how features like bedrooms, area, renovation, and location influence housing prices.
4	Social Impact / Customer Satisfaction	This solution enables real estate buyers, sellers, agents, and market researchers to make informed decisions. It improves housing transparency, supports better urban planning, and enhances user engagement with clear visuals and actionable insights.
5	Business Model (Revenue Model)	This dashboard can be scaled and offered as a subscription-based SaaS tool to real estate companies, market research firms, or housing consultancies. Advanced forecasting modules, API integrations, and custom dashboards can be monetized as premium features.
6	Scalability of the Solution	The system is designed to be scalable and adaptable. It can incorporate new datasets (like rental trends or economic indicators), extend to new regions or cities, and integrate with ML models for price predictions, thereby offering long-term growth potential.

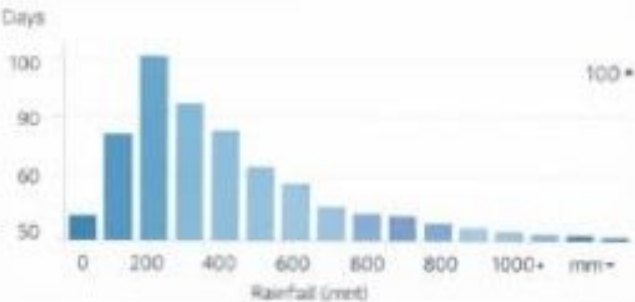
Exploratory Analysis of Rainfall Data in India for Agriculture

Data Analysis Dashboard

State-wise Rainfall Overview



Rainfall Distribution



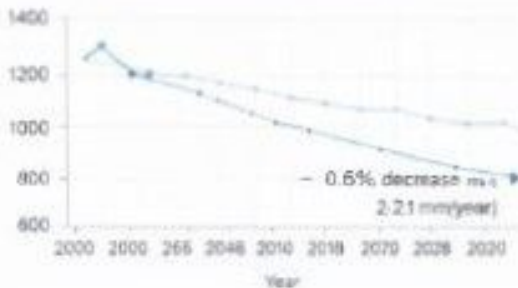
Top 5 Rainiest States



Monthly Rainfall Pattern



Yearly Rainfall Trend in India



Insights and Key Findings

- ✓ North India receives highest rainfall in July
- ✓ West India shows declining trend
- ✓ East India has consistent moderate rainfall



Story Layout

Now story port

Blank Duplicate

Sheet 1

Sheet 2

Sheet 3

Sheet 4

Sheet 5

Sheet 6

Sheet 7

Dashboard 1

A. Drag to add text

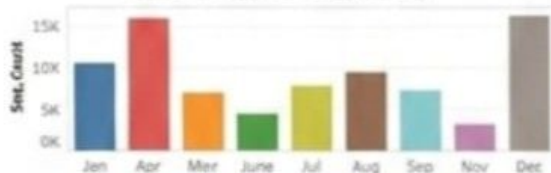
Show title

Size

Story (1018 x 964)

Exploratory Analysis of Rainfall Data in India for Agriculture

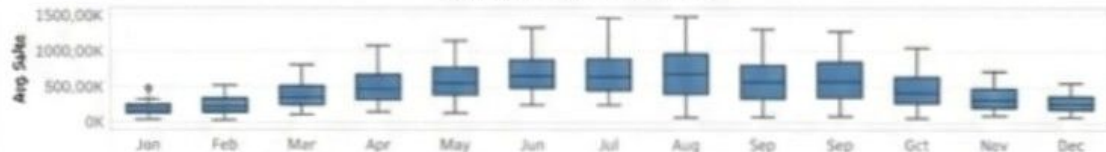
Seasonal Rainfall Distribution



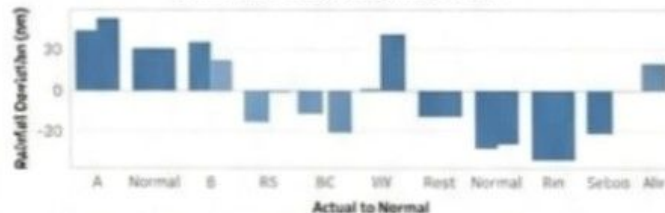
State-wise Peak Rainfall Trends



Monthly Rainfall Variability



Rainfall Deviation from Normal



Impact Analysis on Crop Yield

